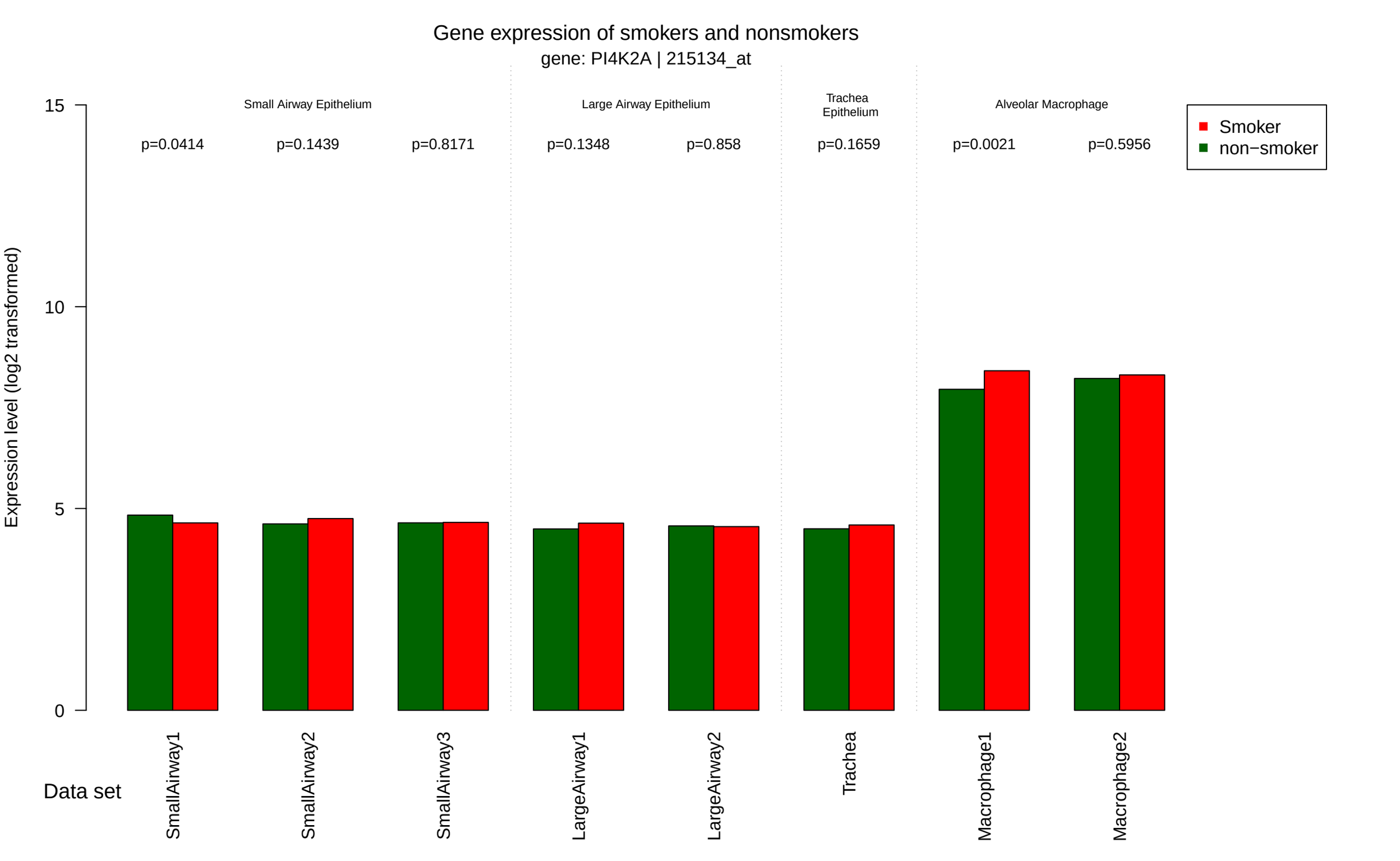
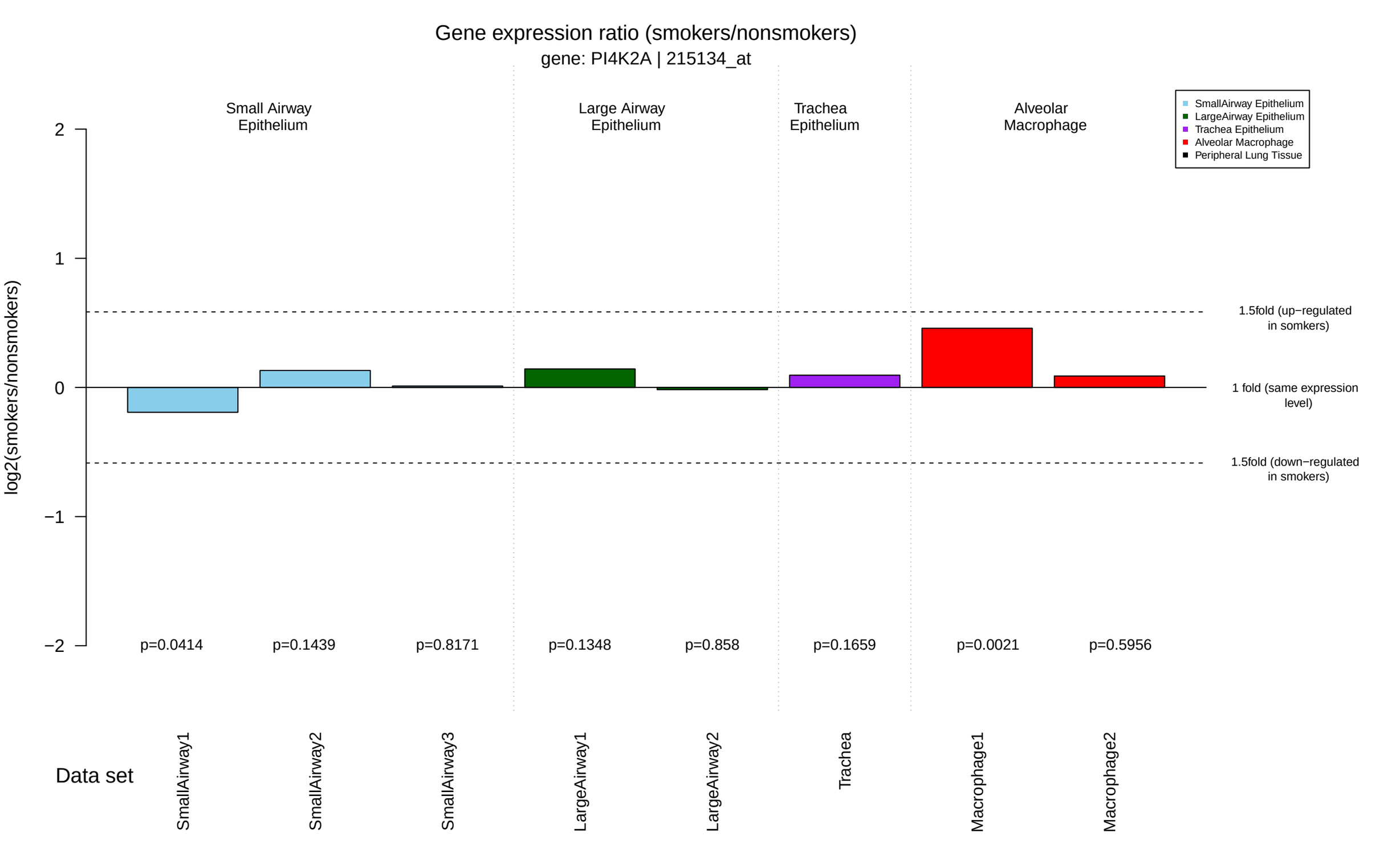
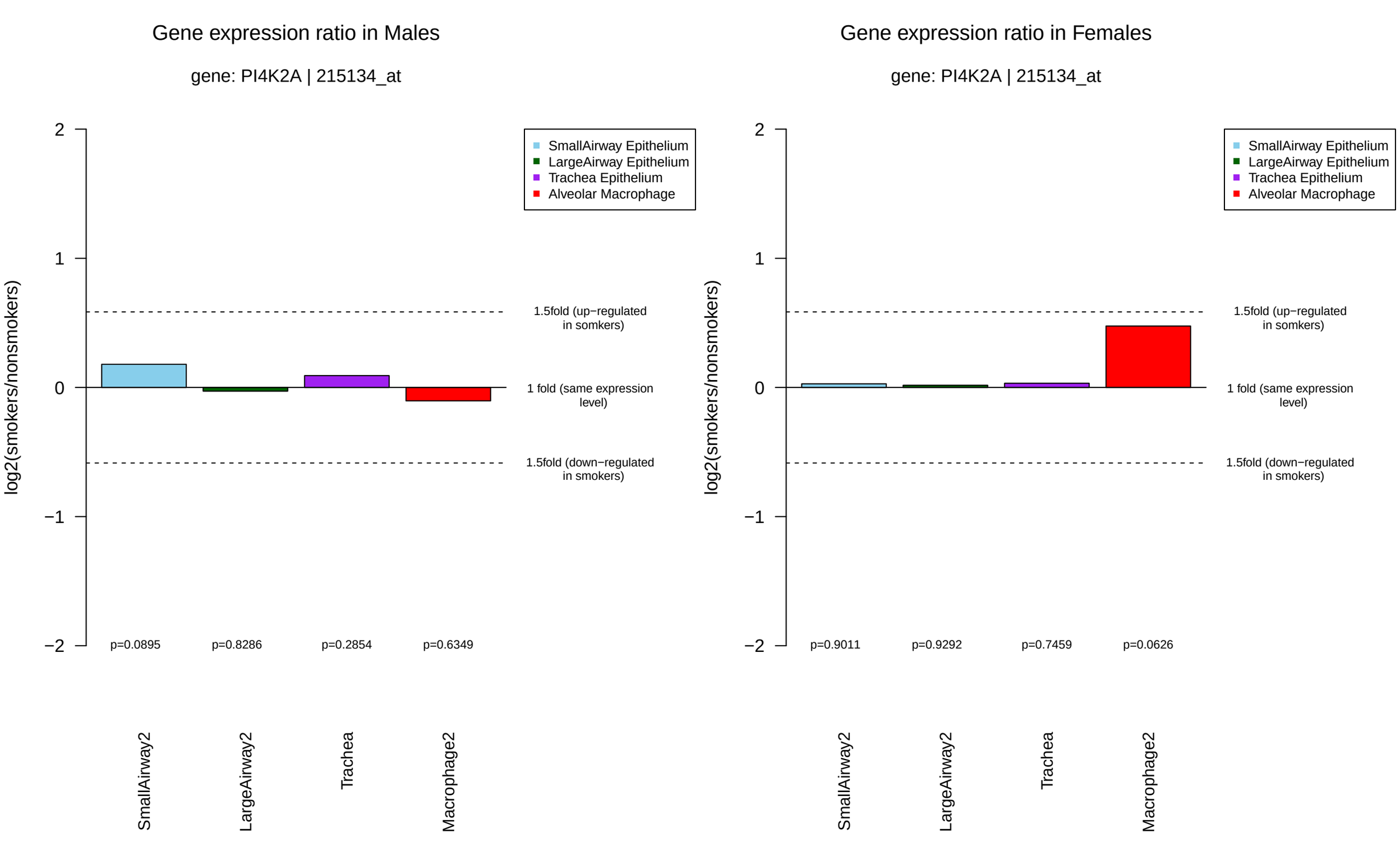
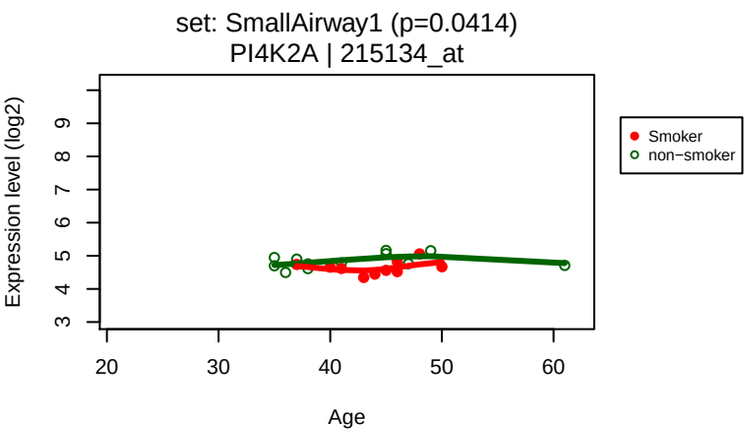
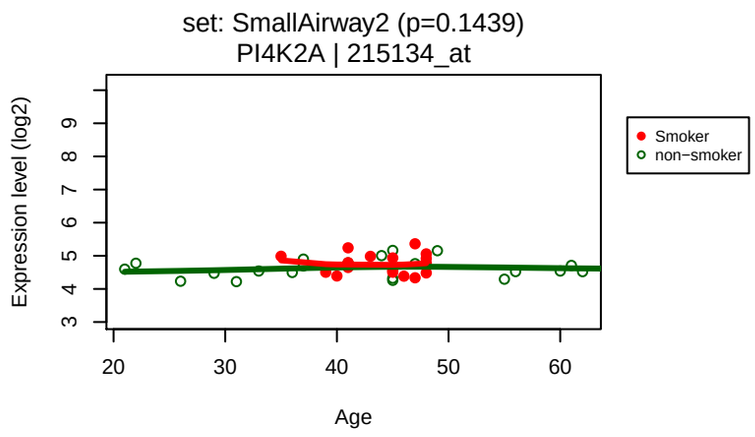
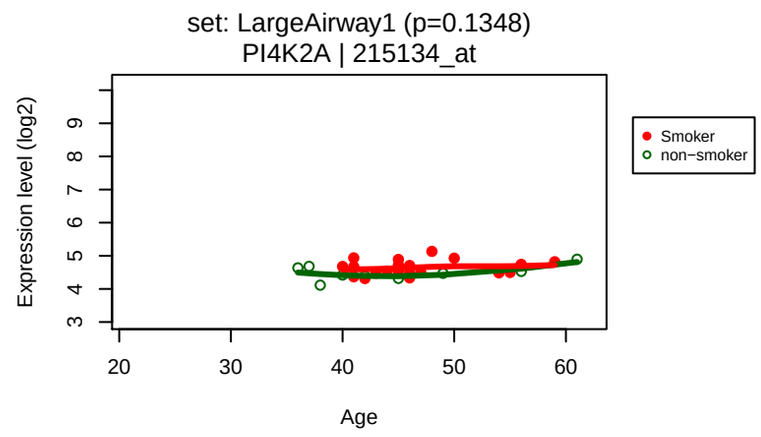

Gene expression of smokers and nonsmokers
gene: PI4K2A | 215134_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0414
p=0.1439
p=0.8171
p=0.1348
p=0.858
p=0.1659
p=0.0021
p=0.5956
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

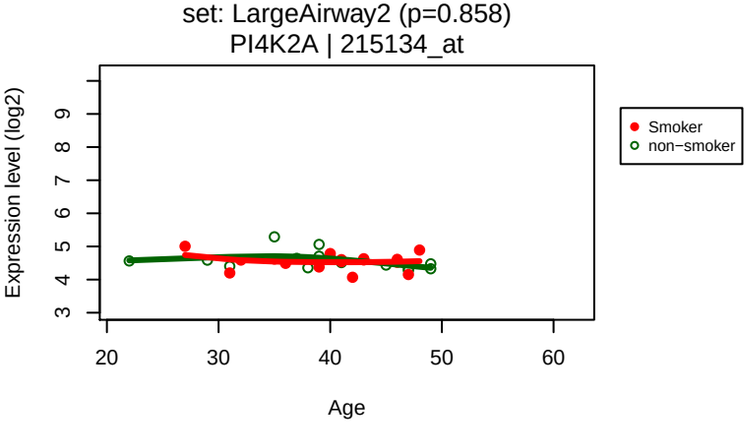

Gene expression ratio (smokers/nonsmokers)
gene: PI4K2A | 215134_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0414
p=0.1439
p=0.8171
p=0.1348
p=0.858
p=0.1659
p=0.0021
p=0.5956
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

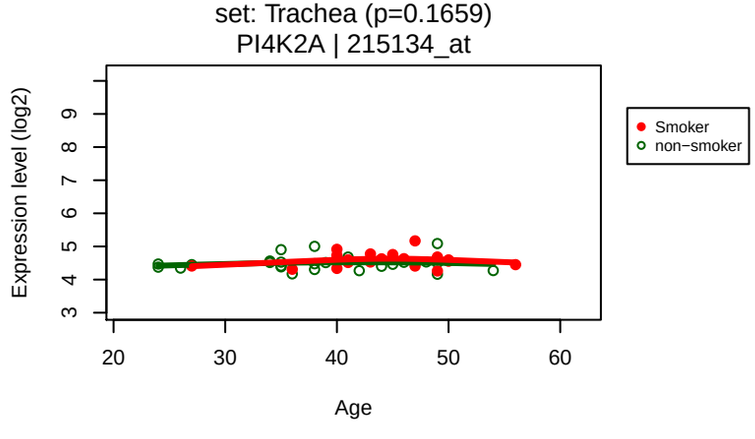

Gene expression ratio in Males
gene: PI4K2A | 215134_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0895
p=0.8286
p=0.2854
p=0.6349
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PI4K2A | 215134_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9011
p=0.9292
p=0.7459
p=0.0626
SmallAirway2
LargeAirway2
Trachea
Macrophage2

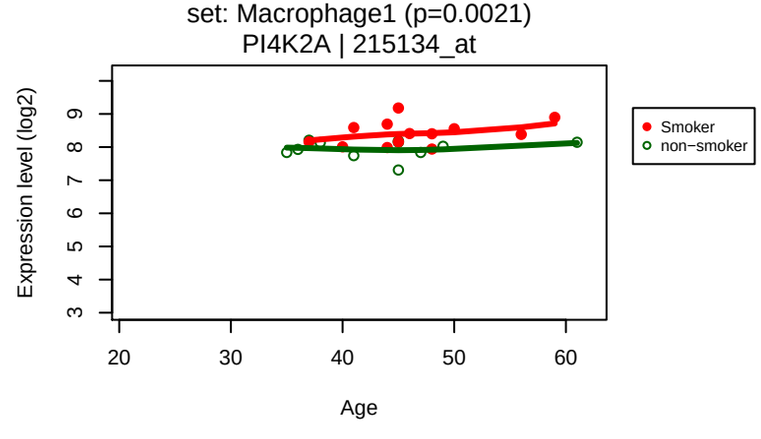

set: SmallAirway1 (p=0.0414)
PI4K2A | 215134_at
Expression level (log2)
Age
Smoker
non−smoker

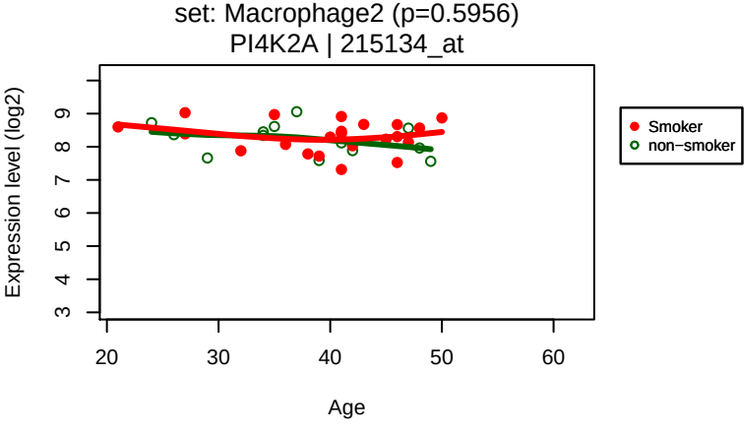

set: SmallAirway2 (p=0.1439)
PI4K2A | 215134_at
Expression level (log2)
Age
Smoker
non−smoker

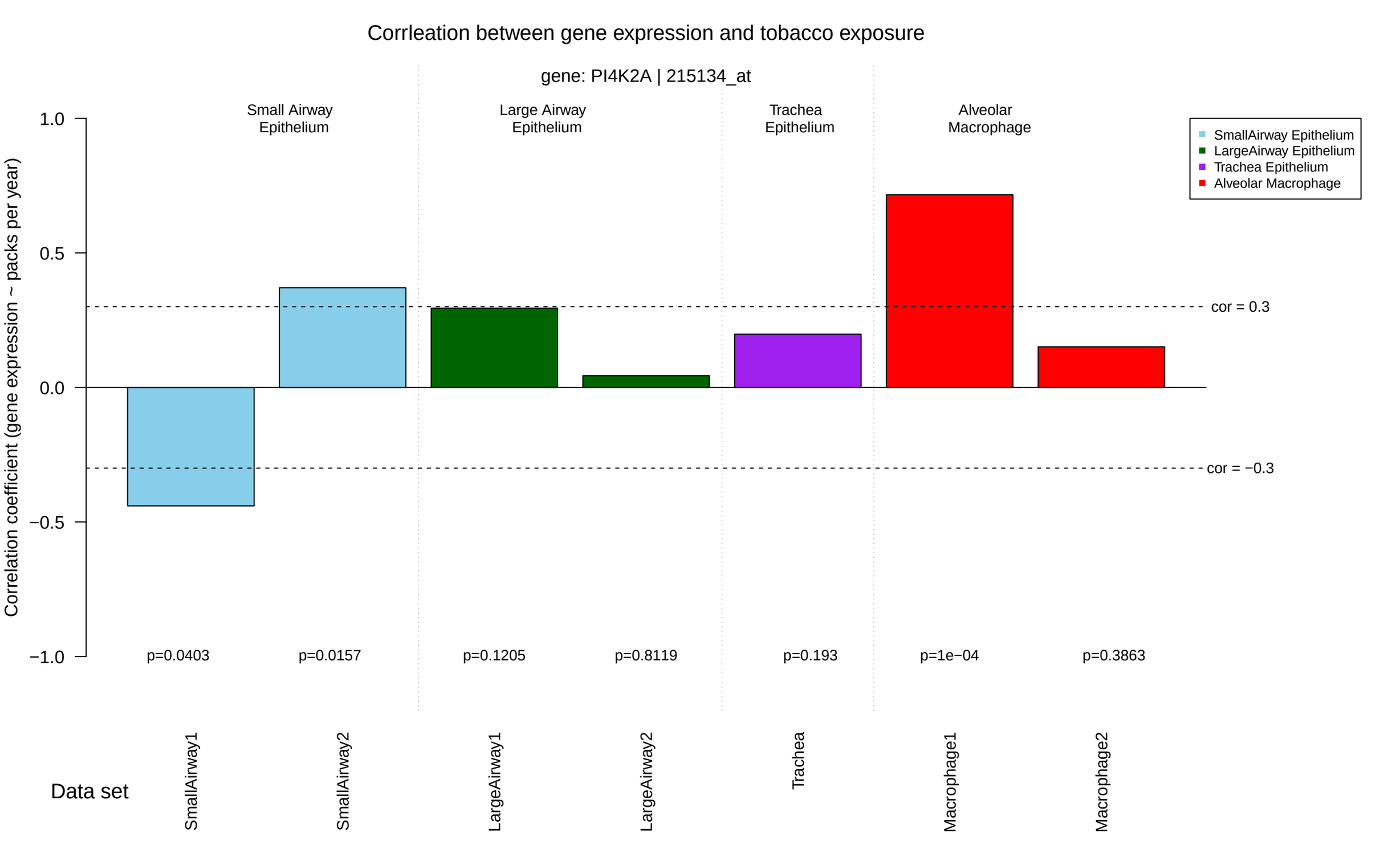

set: LargeAirway1 (p=0.1348)
PI4K2A | 215134_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.858)
PI4K2A | 215134_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1659)
PI4K2A | 215134_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0021)
PI4K2A | 215134_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5956)
PI4K2A | 215134_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PI4K2A | 215134_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0403
p=0.0157
p=0.1205
p=0.8119
p=0.193
p=1e−04
p=0.3863
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2